HW 1 - Zeshui Song

2.16

16. The total charge stored on a 1 cm diameter insulating plate is -10^{13} C. (*a*) How many electrons are on the plate? (*b*) What is the areal density of electrons (number of electrons per square meter)? (*c*) If additional electrons are added to the plate from an external source at the rate of 10^6 electrons per second, what is the magnitude of the current flowing between the source and the plate?

(a)

$$[-10^{13}\,\mathrm{C}]\left[\frac{\mathrm{e}}{1.602\times 10^{-19}\,\mathrm{C}}\right] = -6.242\times 10^{31}\,\mathrm{e} \implies \boxed{6.242\times 10^{31}\text{ electrons}}$$

(b)

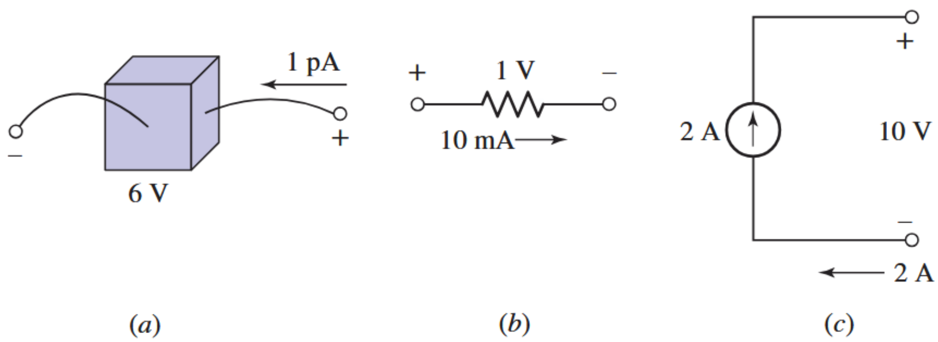
$$\sigma = \frac{6.242\times 10^{31}\text{ electrons}}{\frac{1}{4}\pi(10^{-2}\,\mathrm{m})^2} = \boxed{7.947\times 10^{35}\text{ electrons/m}^2}$$

(c)

$$i = \frac{dq}{dt} = \frac{[10^6\text{ electrons}][1.602\times 10^{-19}\,\mathrm{C/electron}]}{1\,\mathrm{s}} = \boxed{1.602\times 10^{-13}\,\mathrm{C/s}}$$

2.25

25. Determine the power absorbed by each of the elements in Fig. 2.30.



(a)

$$P = [6\,\mathrm{V}][1\times 10^{-12}\,\mathrm{A}] = \boxed{6\times 10^{-12}\,\mathrm{W}}$$

(b)

$$P = [1\,\mathrm{V}][10\times 10^{-3}\,\mathrm{A}] = \boxed{0.01\,\mathrm{W}}$$

(c) The element is supplying 20 W of power.

$$P = [10\,\mathrm{V}][-2\,\mathrm{A}] = \boxed{-20\,\mathrm{W}}$$

<u>Note:</u> The sign convention is that if current enters the positive terminal, then power is being delivered or absorbed. If current enters the negative terminal, then power is being generated or supplied. Also, power is given by $P = VI$.

2.30

30. The current–voltage characteristic of a silicon solar cell exposed to direct sunlight at noon in Florida during midsummer is given in Fig. 2.34. It is obtained by placing different-sized resistors across the two terminals of the device and measuring the resulting currents and voltages. (*a*) What is the value of the short-circuit current? (*b*) What is the value of the voltage at open circuit? (*c*) Estimate the maximum power that can be obtained from the device.

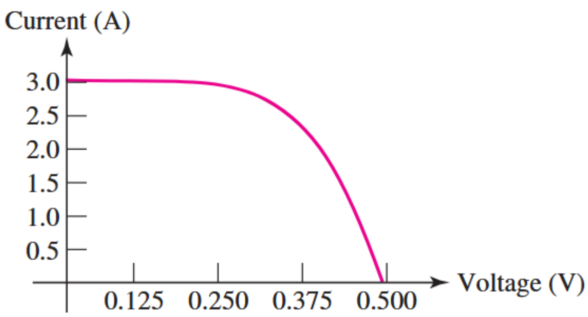


FIGURE 2.34

(a) $\boxed{3.0\,\text{A (short circuit)}}$

(b) $\boxed{0.500\,\text{V (open circuit)}}$

(c) For maximum power, the product of current and voltage is max, this seems to happen around 0.375 V.

$$P_{max} = [0.375\,\text{V}][2.5\,\text{A}] = \boxed{0.9375\,\text{W}}$$

Note: A **short circuit** is defined as a wire with zero resistance, and thus corresponds to a voltage of 0 V across its terminals. An **open circuit** is defined as a break in the circuit with infinite resistance, and thus corresponds to a current of 0 A through its terminals.

2.35

35. A portable music player requiring 5 W is powered by a 3.7 V Li-ion battery with capacity of 4000 mAh. The battery can be charged by a charger providing a current of 2 A with an efficiency of 80%. (*a*) How long can the music player run on a full battery charge? (*b*) How long would it take to fully charge the battery?

(a)

$$w = [3.7\,\text{W}][4000 \times 10^{-3}\,\text{Ah}] = 14.8\,\text{Wh}$$

$$t = \frac{[14.8\,\text{Wh}]}{[5\,\text{W}]} = \boxed{2.96\,\text{h}}$$

(b)

$$I_{eff} = [2\,\text{A}][0.8] = 1.6\,\text{A}$$

$$t = \frac{[4000 \times 10^{-3}\,\text{Ah}]}{[1.6\,\text{A}]} = \boxed{2.5\,\text{h}}$$

Note: Recall that work is the product of power and time, thus energy is given by $\boxed{1\,\text{J} = 1\,\text{W} \times 1\,\text{s}}$. Energy is usually measured in **watt hours (Wh)** where $\boxed{1\,\text{Wh} = 3600\,\text{J}}$.

Battery capacity (energy stored) is then also given in terms of Wh. Since the voltage of a battery is constant, we can define the energy stored in terms of the charge capacity Q (in Ampere hours, Ah) and voltage V (in Volts, V) as:

$$w = \int P\,dt = \int VI\,dt = V\int I\,dt = VQ$$

where Q is in Ah, V is in V, and w is in Wh.

To find how long the battery will last, we can divide its battery capacity by the load/output power.

$$t = \frac{w}{P}$$

To find how long it will take to charge the battery, we can divide its charge capacity by the charging current.

$$t = \frac{Q}{I_{eff}}$$

Where I_{eff} is the effective charging current accounting for any losses ($I_{eff} = I_{in} \times \eta$ where η is the efficiency of the charger).

2.40 40. The circuit depicted in Fig. 2.36 contains a dependent current source; the magnitude and direction of the current it supplies are directly determined by the voltage labeled v_1. Note that therefore $i_2 = -3\,v_1$. Determine the voltage v_1 if $v_2 = 33\,i_2$ and $i_2 = 100$ mA.

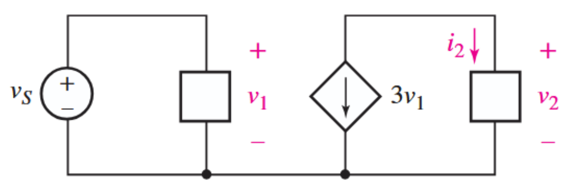


FIGURE 2.36

2.43 43. Real resistors can only be manufactured to a specific tolerance, so in effect the value of the resistance is uncertain. For example, a 1 Ω resistor specified as 5% tolerance could in practice be found to have a value anywhere in the range of 0.95 to 1.05 Ω. Calculate the voltage across a 2.2 kΩ 10% tolerance resistor if the current flowing through the element is (*a*) 1 mA; (*b*) 4 sin 44*t* mA.

2.60 60. A 250 ft long span separates a dc power supply from a lamp which draws 25 A of current. If 14 AWG wire is used (note that two wires are needed for a total of 500 ft), calculate the amount of power wasted in the wire.

3.5

5. Refer to the circuit of Fig. 3.48, and answer the following:
   (a) How many distinct nodes are contained in the circuit?
   (b) How many elements are contained in the circuit?
   (c) How many branches does the circuit have?
   (d) Determine if each of the following represents a path, a loop, both, or neither:
      i) A to B
      ii) B to D to C to E
      iii) C to E to D to B to A to C
      iv) C to D to B to A to C to E

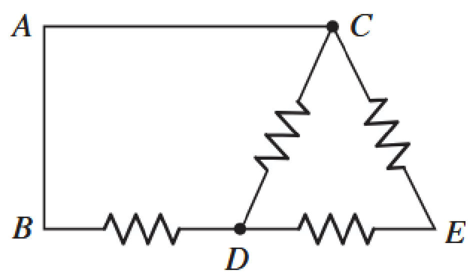


FIGURE 3.48

3.10

10. The circuit of Fig. 3.53 represents a system comprised of an LED sign powered by a combination of battery storage and three solar panels. The panels are not equally illuminated, so the current each supplies can vary, although the voltage

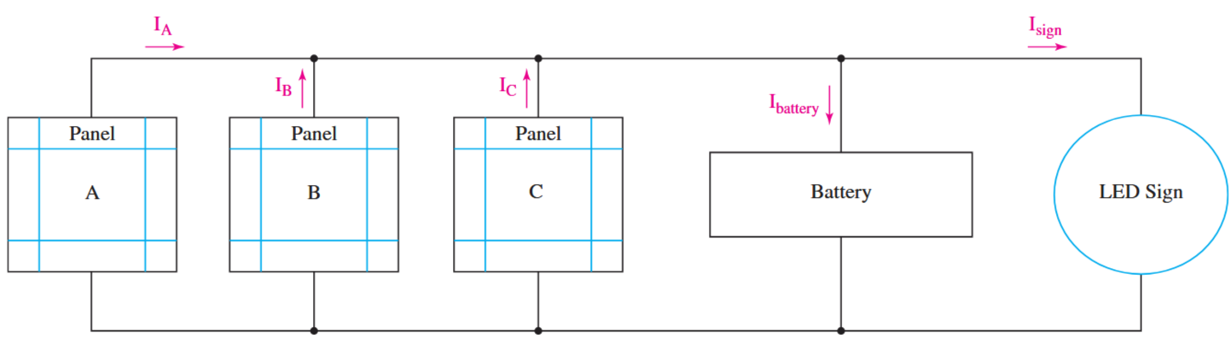


FIGURE 3.53

3.12 12. For the circuit of Fig. 3.55 (which employs a model for the dc operation of a bipolar junction transistor biased in active region), I_C is measured to be 1.5 mA. Calculate I_B and I_E.

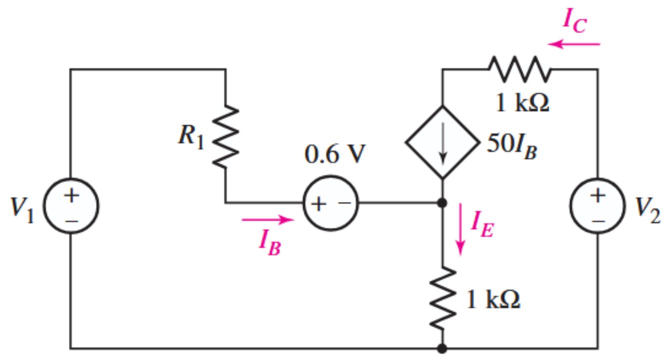


FIGURE 3.55

3.20 20. In the circuit of Fig. 3.55, calculate the voltage across the dependent source ("+" reference on top) if V_2 is 15 V, and I_B is 20 μA.

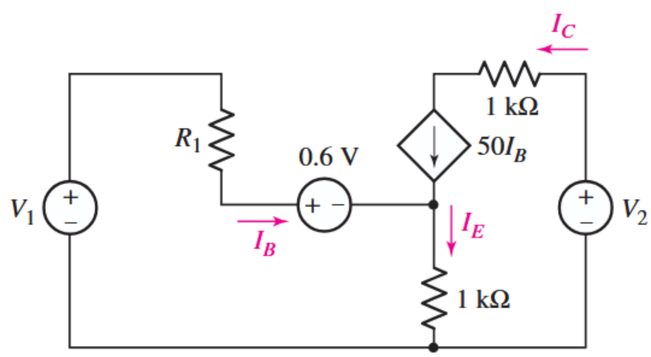


FIGURE 3.55